

RAK14012 WisBlock LED Matrix Datasheet

Overview

Description

RAK14012 is an LED matrix driver module for WS2812B LEDs, which has a control circuit and RGB chip.

To drive the WS2812B, the RAK14012 has a built-in boost circuit to generate 5 V voltage output. But it is recommended to use an external power supply to power the LED Matrix.

Features

- LED matrix driver
- 3.3 V Power supply
- Module Size: 25 mm x 15 mm

Specifications

Overview

Mounting

The RAK14012 WisBlock LED Matrix can be mounted to the IO slot of the WisBlock Base board. **Figure 1** shows the mounting mechanism of the RAK14012 on a WisBlock Base board, such as the [RAK5005-O](#) .

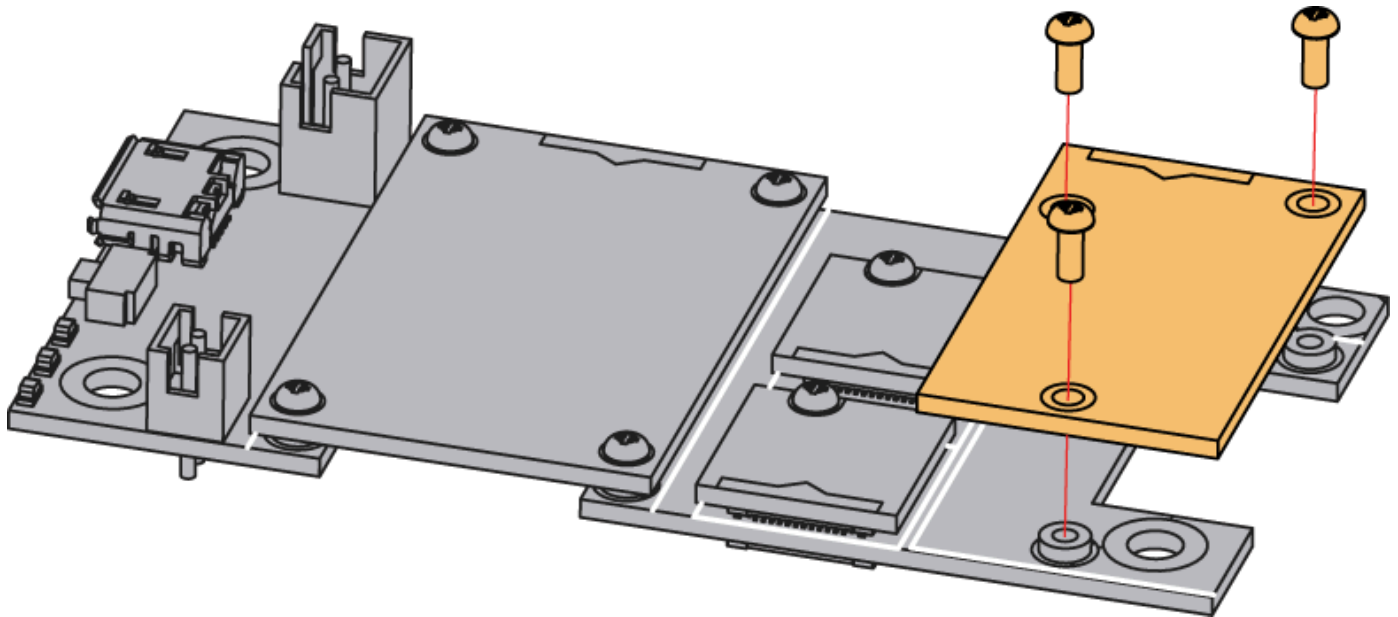


Figure 1: RAK14012 WisBlock LED Matrix mounting

Hardware

Pin Definition

The RAK14012 WisBlock LED Matrix comprises a 40-pin WisConnector. The 40-pin WisConnector allows the RAK14012 module to be mounted to a WisBlock baseboard, such as RAK5005-O. The pin order of the connector and the pinout definition is shown in **Figure 2**.

NOTE

- The **VBAT**, **GND**, **EN (WB_IO6)**, and **DOUT (WB_IO5)** are connected to WisConnector.

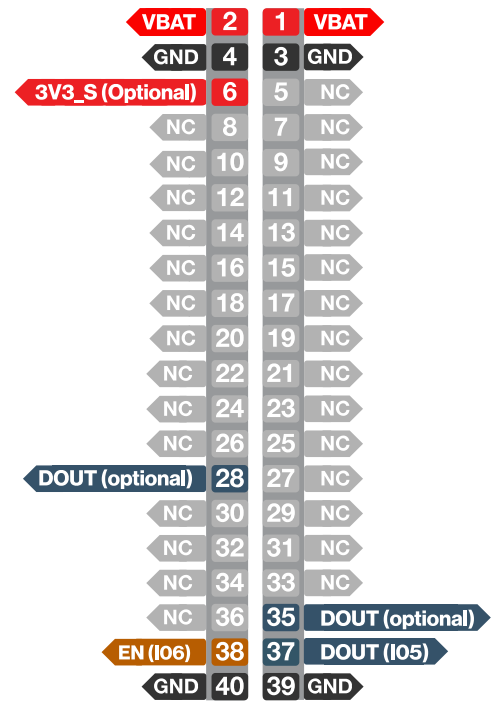
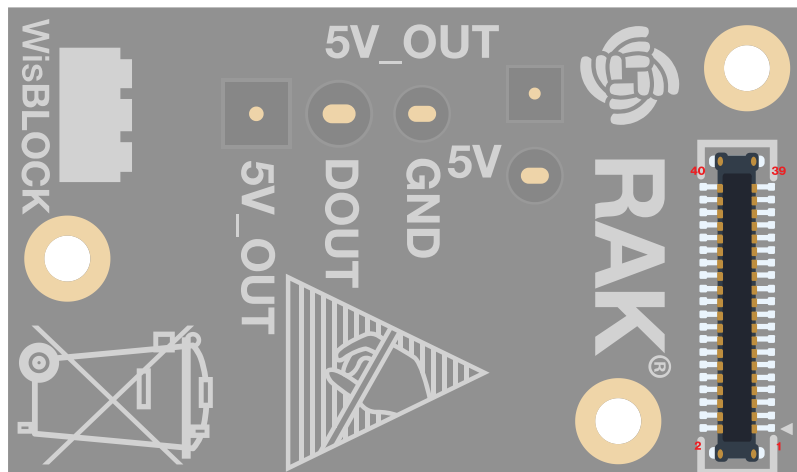


Figure 2: RAK14012 WisBlock LED Matrix pinout

Electrical Characteristics

Symbol	Description	Condition	Min.	Nom.	Max.	Unit
VBAT	Lithium battery output	Normal operation	3.3	-	4.2	V
3V3_S	3.3 V from WisBase	Normal operation	-	3.3	-	V
5V	Output from boost	Normal operation	-	5	-	V

Mechanical Characteristics

Board Dimensions

Figure 3 shows the dimensions and the mechanical drawing of the RAK14012 module.

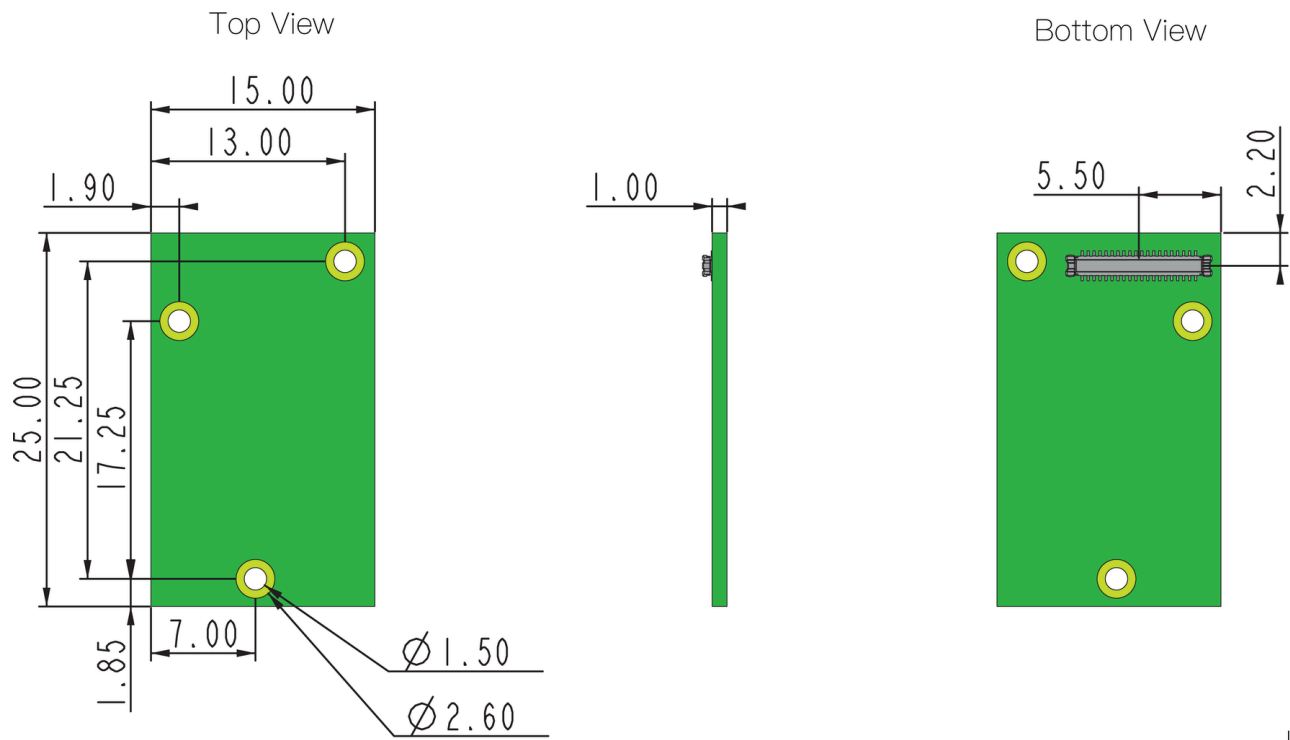


Figure 3: RAK14012 WisBlock LED Matrix dimensions

WisConnector PCB Layout

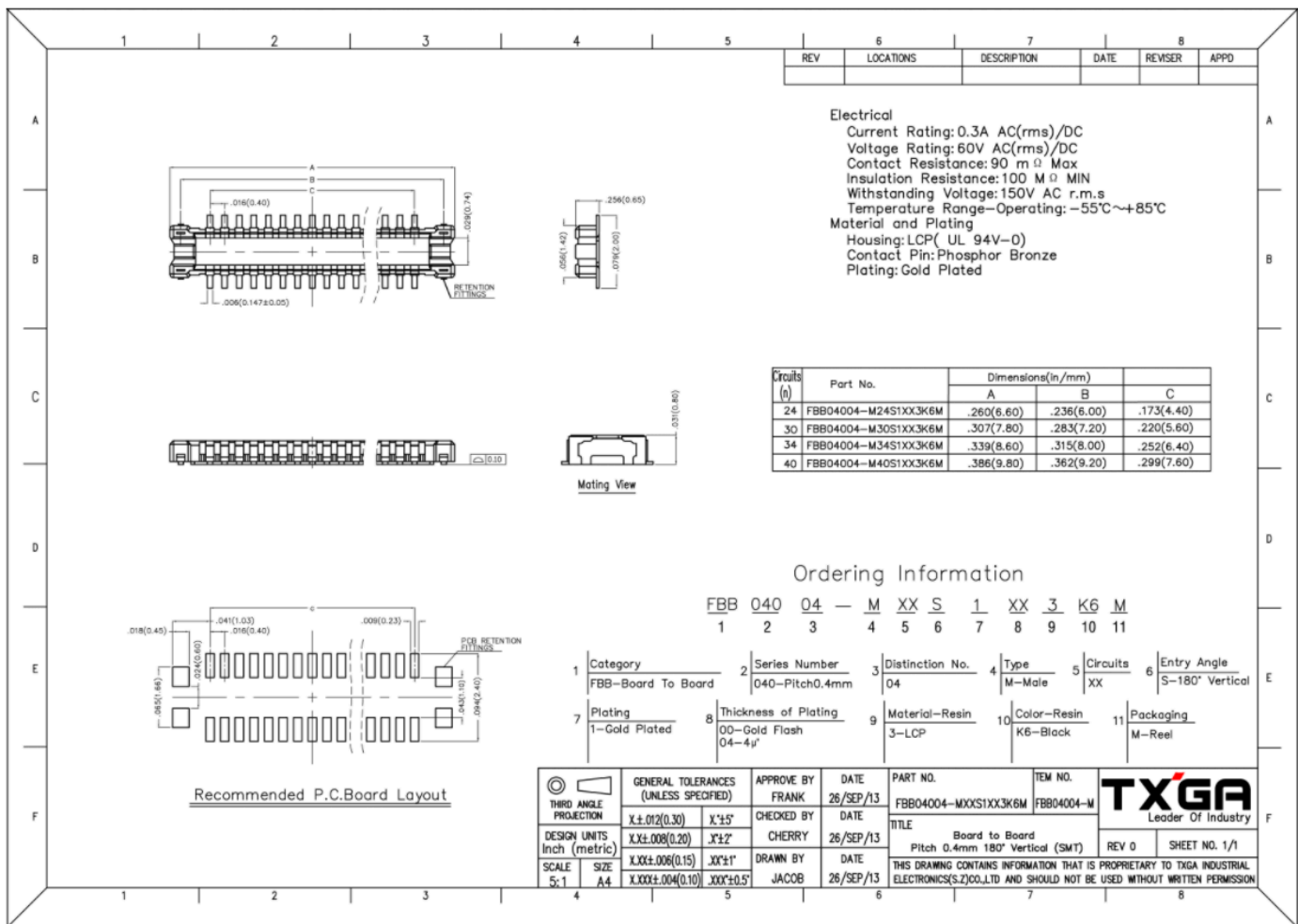


Figure 4: WisConnector PCB footprint and recommendations

Schematic Diagram

It has a boost circuit to generate 5 V because the supply for WS2812B must be between 3.7 V and 5.3 V. It is recommended to use an external power source to power the LED Matrix because of its high power consumption.

Connector

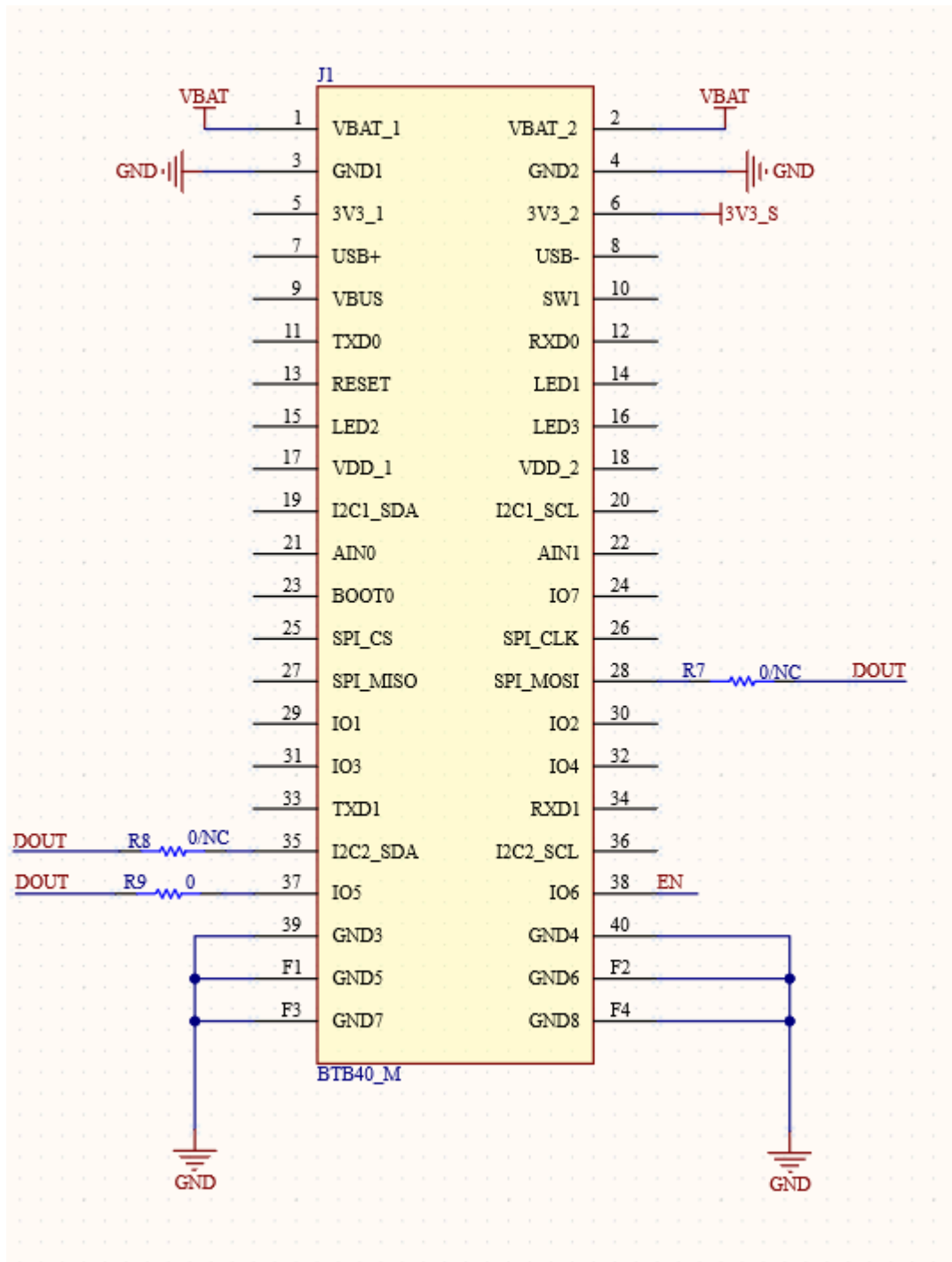


Figure 5: RAK14012 connector connection

Boost

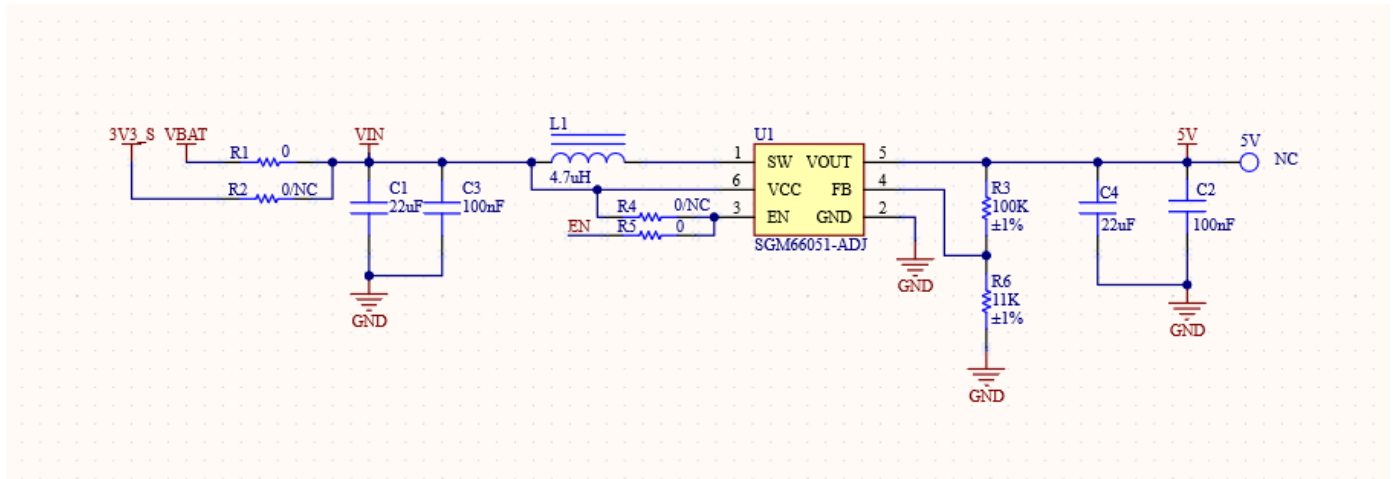


Figure 6: RAK14012 boost circuit

Power Selection

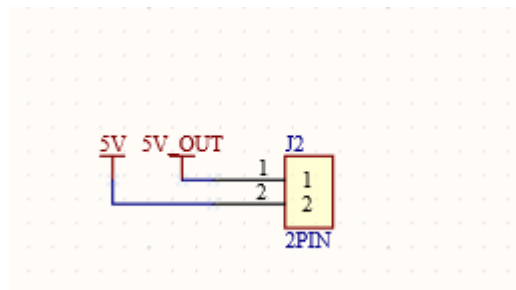


Figure 7: RAK14012 power selection

⚠ WARNING

- Make sure that the jumper from 5V_OUT of RAK14012 WisBlock LED Matrix is disconnected when connecting to an external power supply.
- LED Matrix DIN should be connected to RAK14012 DOUT.
- LED Matrix GND should be connected to RAK14012 GND.

Connector for LED Matrix

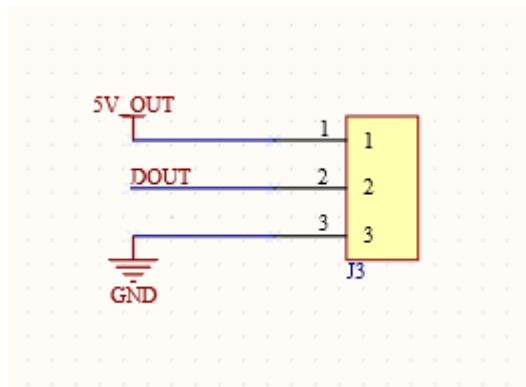


Figure 8: RAK14012 connector for LED matrix